

m6AConquer: a consistently quantified and orthogonally validated database for the *N*⁶-methyladenosine (m⁶A) epitranscriptome

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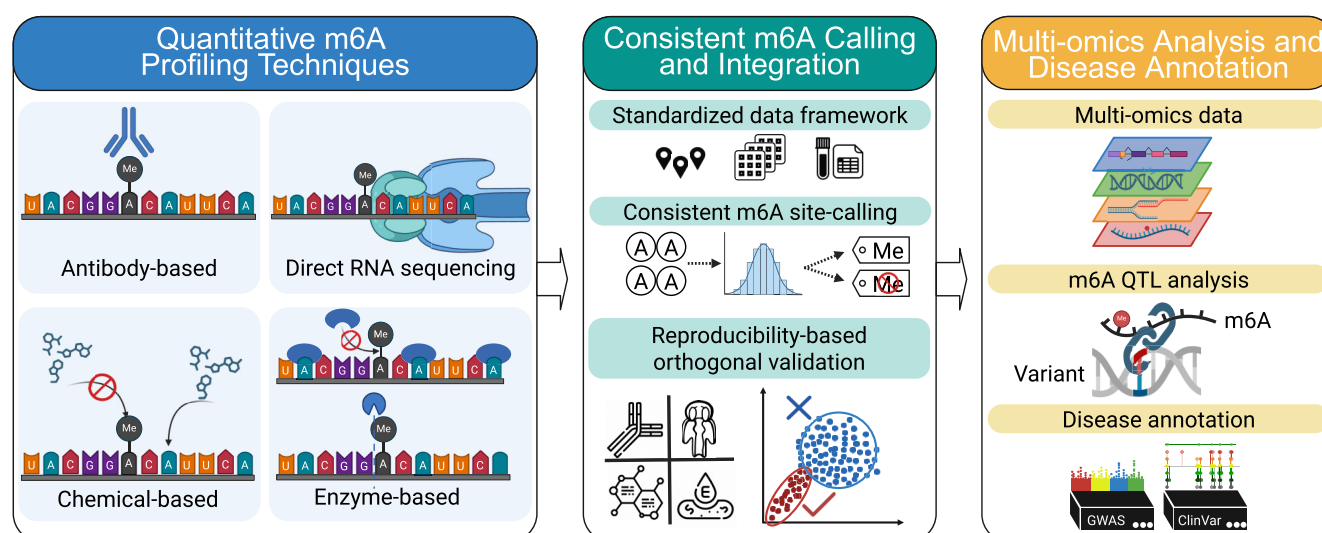
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Abstract

The proper placement of *N*⁶-methyladenosine (m⁶A) on mRNA is essential for normal cell function, and its disruption is linked to numerous human diseases. The rapid growth of m⁶A data from diverse sequencing technologies presents challenges for integrative analysis due to technique-specific biases and inconsistent processing. While existing databases provide valuable catalogs, their reliance on aggregating pre-processed results can propagate inconsistencies. To overcome these limitations, we present m6AConquer, a database founded on reproducible quantification. We systematically re-processed raw data from 10 distinct profiling methods, including high-resolution GLORI and eTAM-seq, quantifying methylation at millions of consensus sites across human and mouse. Our rigorous pipeline features uniform site-calling and false-positive calibration with in vitro transcribed (IVT) controls where available. By leveraging a reproducibility-based framework across technically orthogonal methods, we identified over 135300 orthogonally validated m⁶A sites in human (IDR < 0.05). Beyond this validated methylome, m6AConquer provides matched multi-omics data (gene expression, splicing, variants) and identifies m⁶A quantitative trait loci (m⁶A QTLs) to link RNA modification to genetic regulation and disease. Offering intuitive query tools, interactive visualizations, and downloadable, analysis-ready data matrices, m6AConquer provides a standardized resource for rigorous exploration of the roles of m⁶A in biology and medicine, freely accessible at <https://rnamd.org/m6aconquer/>.

Graphical abstract



Introduction

*N*⁶-methyladenosine (m⁶A) modification is one of the most widely studied RNA chemical modifications in eukaryotic

cells [1]. m⁶A epitranscriptomically regulates gene expression via specifying RNA half-life [2–8]. It is the most abundant type of RNA modification, topologically and functionally similar

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to DNA 5mC modification [9]. The basal constitutional level of m⁶A on transcripts functions as a negative gene expression regulation mechanism by targeting the modified RNA to P-bodies for degradation [1, 4, 5, 9, 10]. The installation of m⁶A on transcripts is mainly dictated by the DRACH consensus motif, but it is suppressed in regions close to splicing junctions [11]. This mechanism makes the intron-exon architecture a key determinant of RNA half-life, with longer exons harboring more m⁶A modifications [12–14].

m⁶A modification state is dynamically regulated by protein factors, including writers (*METTL3*, *METTL14*, *WTAP*, *KIAA1429*), erasers (*FTO*, *ALKBH5*), and readers (*YTHDF1-3*, *YTHDC1-2*, *HNRNPC*) [6, 15–23]. The specific m⁶A regulators allow the same m⁶A site link to distinct downstream outcomes via alternative mechanisms. m⁶A can also interact with epigenetic modules, including histone methylation (H3K36me3) and transposable element (SINEs and LINEs) expression [24–30]. m⁶A on noncoding RNAs, such as miRNAs and carRNAs, has also been reported to influence their processing and function. Furthermore, m⁶A plays a crucial role in fundamental cellular processes such as the cell cycle, differentiation, and stress responses [4, 31–36]. Accordingly, m⁶A is known to be involved in the development of disease and other medical conditions, including acute myelogenous leukemia (AML) [37–39], depression [40–42], and multiple cancers [43–46]. Perturbation of m⁶A regulators, like *FTO*, can also impact crop yields, as observed in *maize* [47–49].

Existing sequencing techniques for profiling the m⁶A methylome fall into four main groups: antibody-assisted, enzyme-assisted, chemical-assisted, and direct RNA sequencing methods. Antibody-based approaches, such as MeRIP-seq [15, 50] and miCLIP [51], enrich methylated RNA fragments through immunoprecipitation. Enzyme-assisted methods like eTAM-seq [52] and MAZTER-seq [53] use site-specific enzymatic cleavage or base-conversion reactions. Chemical-assisted protocols, including GLORI [54], convert unmethylated adenosines to inosines, facilitating quantitative discrimination. Lastly, direct RNA sequencing leverages distinct ionic current patterns generated when m⁶A-modified nucleotides pass through nanopores [55–58].

Despite their contributions to transcriptome-wide m⁶A mapping, each method has specific limitations. Antibody-based techniques frequently exhibit high false-positive rates (up to ~30%), mainly because the antibody binds unmodified RNA regions whose sequence motifs resemble those typically found near true m⁶A-modified sites [59]. The high coverage of these regions in highly expressed transcripts further increases the likelihood of false-positive calls [59, 60]. Enzyme-assisted approaches often experience incomplete or inaccurate enzymatic reactions, exemplified by false-positive rates exceeding 50% in MAZTER-seq [59, 60]. Direct RNA sequencing continues to face challenges from limited IVT-generated training data and overlapping current signatures, complicating accurate site detection [59, 61]. Similarly, chemical-assisted methods like GLORI can encounter issues with incomplete conversions influenced by RNA secondary structures [62, 63].

Consequently, most existing methods lack reliable absolute quantification and exhibit inherent limitations in site resolution, making their measurements imprecise and requiring orthogonal validation to accurately characterize the transcriptome-wide distribution and dynamics of m⁶A methylation. In recent studies, technically independent validations have been increasingly adopted, where intersections between

techniques reinforce the understanding of the true m⁶A installation mechanism in the transcriptome [12, 60, 64, 65]. For example, using common sites from miCLIP, m⁶A-SAC-seq, DART-seq, and GLORI, Anna *et al.* show that m⁶A formation is driven by the splice-site exclusion mechanism [12, 13]. Another study using a similar integrative analysis found that m⁶A does not promote protein translation [65].

While several valuable databases, including m⁶A-Atlas2, MeT-DB, RMBase2, REPIC, DirectRMDb, and Sci-ModoM, have emerged to catalogue m⁶A sites [66–71], the field still grapples with a fundamental challenge: the lack of a standardized and validated data-sharing framework for integrative analysis. Current databases typically act as data aggregators, compiling pre-computed peaks or modification sites from individual publications. This approach, however, inherits and propagates critical limitations. Firstly, it results in inconsistent site reporting, as different studies employ different peak-calling algorithms and statistical thresholds, making direct comparisons across datasets unreliable [59]. Secondly, these databases almost exclusively report “positive” sites, neglecting the vast landscape of high-coverage, low-methylated or unmethylated adenosine loci, which are essential for building generalizable predictive models and understanding regulatory specificity. This focus on pre-identified positive sites, common in many current resources, leads to a biased view of the methylome.

To address these critical challenges, we introduce a reproducibility-based integration strategy that moves beyond conventional data aggregation. Here, we present m6AConquer, a resource built upon this principle, which introduces three key advances:

1. **Consistent quantification:** We established a unified quantification strategy by processing all raw data from scratch. Instead of aggregating disparate published peak lists, we systematically re-quantified data from ten distinct profiling technologies against a unified set of over 5.5 million standardized loci across human and mouse. This process ensures two key advantages: first, every sample is measured against the same set of loci, enabling direct and reliable comparisons; second, it generates complete, matrix-based quantifications for all sites, eliminating the missing values inherent in peak-only databases. This homogeneous quantification is further enhanced by a uniform site-calling procedure and empirical false-positive calibration using *in vitro* transcribed (IVT) controls.
2. **Orthogonal validation:** A core improvement of our database is an integration strategy based on the Irreproducible Discovery Rate (IDR), a well-established standard in ENCODE projects [72]. We adapted this statistical framework to assess the quantitative agreement of methylation level (m⁶A ratio) between technically orthogonal methods, i.e. those rooted in distinct biochemical principles (e.g. antibody enrichment versus enzyme-based conversion). By demanding reproducibility across independent techniques, this approach rigorously filters out technology-specific artifacts and moves beyond simple site overlaps, resulting in a reliable set of orthogonally validated m⁶A sites.
3. **Multi-omics integration:** Finally, m6AConquer is the first resource in this field to provide matched, sample-specific multi-omics data, including transcriptomics, al-

ternative splicing, and genetic variants, within a unified and analysis-ready format. This rich integration facilitates direct exploration of m⁶A's functional consequences. As a demonstration, we used the multi-omics quantification to identify m⁶A quantitative trait loci (m⁶A QTLs) on our well-supported set of orthogonally validated sites. These associations can link genetic variants to RNA modification levels and potential disease risk. The QTL-based approach complements existing databases that rely on in-silico mutation effect predictions, which have shown inconsistencies with empirical m⁶A QTL evidence [73]. Key features distinguishing m6AConquer from other databases are summarized in Table 1.

m6AConquer is therefore presented not merely as a database, but as an inclusive data discovery platform. It provides the community with consistently quantified m⁶A signals and a stringently defined set of orthogonally validated sites, all organized within analysis-ready matrix-based data structures. Through an intuitive web interface, users can query, visualize, and download these large-scale and diverse datasets, facilitating reliable and detailed investigation into the mechanisms and functions of RNA methylation. All data, tools, and documentation are freely available at <https://rnamd.org/m6aconquer/>.

Materials and methods

Data collection and preprocessing

Raw sequencing data were collected from 10 quantitative m⁶A profiling technologies, spanning antibody-assisted (MeRIP-seq [66], m6ACE-seq [74, 75]), enzyme-assisted (eTAM-seq [52], m6A-SAC-seq [76, 77], MAZTER-seq [53], DART-seq [78], scDART-seq [79], and m6A-REF-seq [59, 80]), direct RNA sequencing [58], and chemical-assisted (GLORI [54]) methods. In total, we retrieved 1928 sequencing runs (FASTQ/FAST5 files) corresponding to 1393 biological samples from GEO [81, 82], GSA [83, 84], and EMBL-EBI [85], covering both *Homo sapiens* and *Mus musculus*. Adapter sequences and low-quality reads were removed using Cutadapt (v.2.9) [86], Trim Galore (v.0.6.6) (<https://github.com/FelixKrueger/TrimGalore>), and Flexbar 3.0 [87]. PCR duplicates were eliminated via Samtools (v.1.10) [88], FASTX-Toolkit (v.0.0.13) (https://github.com/agordon/fastx_toolkit), and UMI-tools (v.1.1.4) [89]. Fastp (v.0.23.4) [90] performed both trimming and deduplication. For NGS sequencing data, read alignment was performed with HISAT2 (v.2.1.0) [91], HISAT-3N (v.2.2.1–3n-0.0.3) [92], or STAR (v.2.7.11a) [93]. Direct RNA sequencing data were base-called using Guppy (v.3.1.5) and aligned via Minimap2 (v.2.17) [94]. Nanopolish (v.0.13.3) [95] was used to index and align signal events, and m6Anet (v.2.1.0) [58] was used to infer m⁶A sites. Reference genomes were consistently obtained from Ensembl, using GRCh38 (hg38) for human and GRCm38 (mm10) for mouse.

Standardization of multi-omics data

m6AConquer consistently quantifies data from diverse profiling techniques against reference feature sets to construct standardized matrices (Fig. 1). The database incorporates four primary data types: m⁶A methylomes, expression levels, alternative splicing events, and variant calling results.

Definition of reference loci

To obtain the methylation quantification matrices, we first defined a broad reference feature set for human (2 788 705 A sites) and mouse (2 906 531 A sites). These sets were constructed by taking the union of all canonical exonic DRACH motifs from the Ensembl v.110 annotation and the non-DRACH/intronic m⁶A sites previously identified by GLORI in the supplementary data of Liu *et al.*'s study(54).

Quantification of m⁶A methylation and transcript expression

All sequencing samples were quantified against the single-based reference feature sets from read alignments without any post-alignment filtering. Methylation signals were consistently summarized using a binomial approach with two measures: (i) m⁶A count: the number of reads capturing methylated adenosines at each locus (m⁶A-specific reads), and (ii) total coverage: the total number of aligned reads from the same sample (or paired input control) representing transcript coverage at that locus. For example, for antibody-assisted techniques (e.g. MeRIP-seq, m6ACE-seq), m⁶A counts were obtained by counting IP read fragments overlapped with the single-based reference A sites using the summarizeOverlaps function from the GenomicAlignments package (v1.40.0). The total coverage was obtained by summing up the counts derived from the IP sample and its paired input sample (counted in the same way as IP). Further details on methylation quantification methods for each sequencing technique are available in the Supplementary Protocol. In addition to the RNA methylome, gene expression levels were quantified from read alignment files of the methylation assays (input samples only for antibody-assisted methods), using Ensembl v.110 gene annotations. RPKM values were computed from raw read counts to normalize the expression levels.

Alternative splicing detection and variant calling

The alternative splicing matrices employed non-overlapping exonic regions from Ensembl v.110 annotation as reference features. rMATS (v.4.3.0) [96] was used to compute the percent-spliced-in (PSI) scores for five splicing event types (skipped exons, alternative 5' splice sites, alternative 3' splice sites, retained introns, and mutually exclusive exons; parameters: –variable-read-length –statoff –individual-counts). SNP data were generated using GATK (v.4.6.1.0) [97] according to the pipeline of the previous study [98]. Variants were called via HaplotypeCaller (–stand-call-conf 20 –minimum-mapping-quality 20) and filtered with VariantFiltration (–filter-name FS –filter “FS > 30.0” –filter-name QD –filter “QD < 2.0”). Only dbSNP-annotated [99] variants, excluding UCSC RepeatMasker microsatellites [100] and REDIPortal database [101] entries, were retained.

Building standardized data-sharing frameworks

The data matrices for all omics modalities were organized as SummarizedExperiment (SE) or RaggedExperiment (RE) objects [102] (RE) with genomic coordinates of reference features (stored in rowRanges) and sample information (stored in colData). For m⁶A SEs, the matrices (accessed through assays()) include m6A counts and total coverage counts for each reference A site (rows) across all samples (columns). These count values have no missing data, making them reliable for

Table 1. Comparison of features across M6aconquer, m6a-atlas, m6a-atlas2, Direct-RMDB, and Sci-ModoM about m⁶A modification on the human genome

Item	m6A-Conquer	m6A-Atlas	m6A-Atlas2	Direct-RMDB	Sci-ModoM
Quantification techniques	10	1	1	1	5
m ⁶ A reference sites	2 788 705	178 049	427 760	195 871	894 098
Samples/cells	1393	920	1481	18	80
Unified site-calling and quantification	✓	–	–	–	–
Orthogonal reproducible integration	✓	–	–	–	–
Standardized matrix datatype	✓	–	–	–	–
Transcriptomics and alternative splicing	✓	✓	✓	✓	–
Sample-specific SNP data	✓	–	–	–	–
m ⁶ A QTL	✓	–	–	–	–

benchmarking alternative models or tool fits. Multi-omics integration was performed using MultiAssayExperiment (MAE) [103] to unify the data objects from each profiling technique, ensuring that sample identifiers are consistent across all datasets. Details of the data-sharing frameworks are provided in the Supplementary Protocol.

IVT calibration and uniform site-calling

After constructing the uniform data-sharing framework, IVT samples were used to calibrate read count matrices for MeRIP-seq, GLORI, and eTAM-seq by masking false-positive sites. IVT samples serve as modification-free negative controls, with gene expression levels matching those of regular mRNA samples [59]. For MeRIP-seq, false positives were identified using the machine learning-based tool m6ACali (v1.0.0) (60), with a false-positive probability threshold set at > 0.5. For GLORI and eTAM-seq, false-positive sites were determined by quantifying IVT samples against the reference m⁶A lites and using a site-calling binomial test with a BH-adjusted *P*-value threshold of < 0.05 (detailed below). To ensure consistency across the dataset, m⁶A-specific read counts and total read coverage at these false-positive sites were set to zero within the data-sharing framework.

After IVT calibration of the read count matrices, we selected the binomial test with Benjamini–Hochberg (BH) correction as the baseline site-calling method for all sequencing techniques. Specifically, we tested for m⁶A enrichment at each site using a one-sided binomial test, comparing the observed m⁶A counts to the expected proportion defined by the average m⁶A modification level across the entire sample (i.e. $\text{sum}(\text{m}^6\text{A}) / \text{sum}(\text{Total})$). This approach is consistent with the strategies employed by GLORI [54] and eTAM-seq [52]. In parallel, we applied the beta-binomial mixture (BBmix) model as a complementary, probabilistic method to classify m⁶A sites. Originally developed for peak-calling in MeRIP-seq data by MeTPeak [104] and subsequently adopted by MeTDB2 database [67], BBmix estimates the posterior probability of each site belonging to the m⁶A-enriched foreground. Compared to relying solely on *P*-values, the posterior probabilities generated by BBmix facilitate a clearer distinction between positive (high m⁶A levels) and negative (low m⁶A levels) modification sites at a classification threshold of $\text{prob m}^6\text{A} > 0.5$. Both the BH-adjusted binomial *P*-values and BBmix posterior probabilities are stored as additional assay matrices in the data-sharing frameworks. To ensure data completeness, no fixed *P*-value or posterior thresholds are applied to the quantification matrices; instead, users must define their own thresholds during data analysis. All scripts for the binomial tests and BBmix modeling are available in

the reproducible R package OmixM6A, freely accessible at <https://github.com/ZW-xjtlu/OmixM6A>.

Reproducibility-based technically orthogonal validation

m6AConquer employs a reproducibility-based approach to integrate quantitative m⁶A profiles across orthogonal sequencing methods. Reproducibility is assessed using the Irreproducible Discovery Rate (IDR) statistical framework in the R package *idr* (v.1.3) [105], which is the standard method for measuring reproducibility in large-scale ENCODE projects [72]. IDR uses a two-dimensional Gaussian copula mixture model to model the agreement between ranks of paired measurements, assigning each site an IDR score that quantifies the probability it belongs to a reproducible signal component rather than noise. The design of IDR is independent of the threshold choice for each marginal sample, making it suitable for integrating NGS experiments across different platforms, as noted in the discussion section of Li *et al.* [105]. In m6AConquer, the input to IDR was the m⁶A modification levels, expressed as beta-values, which are calculated for each reference site within each technique. The beta-value (β) is defined in Equation 1:

$$\beta = \frac{\text{m}^6\text{A}}{\text{m}^6\text{A} + \text{A}} \quad (1)$$

where “m⁶A” refers to the number of m⁶A-specific read coverage, and “A” refers to the unmodified coverage in the corresponding site. The choice of beta-values rather than site-calling *P*-values as input for integration was intended to mitigate confounding by gene expression levels, since highly expressed genes tend to yield smaller *P*-values primarily due to increased coverage rather than proportionally higher m⁶A modification.

To identify orthogonally validated m⁶A sites, integration was performed separately for each pair of techniques classified as technically orthogonal. We defined technical orthogonality as pairs of m⁶A profiling methods originating from distinct biochemical principles: antibody-assisted (MeRIP-seq, m6ACE-seq), enzyme-assisted (MAZTER-seq, eTAM-seq, DART-seq, scDART-seq, m6A-SAC-seq, m6A-REF-seq), chemical-assisted (GLORI), or direct RNA sequencing (ONT-DRS). However, direct RNA sequencing methods and antibody-assisted approaches were considered as non-orthogonal pairs because the m⁶A prediction algorithm used in ONT-DRS (m6Anet(58)) was trained on m6ACE-seq data. Within each technique, m⁶A and total read counts were aggregated across all baseline (non-perturbation) samples to represent the underlying modification state while avoiding bi-

ases introduced by experimental perturbations (such as writer knockdowns). For condition-specific analyses, counts were additionally aggregated separately by cell line or tissue type.

For each pair of techniques subjected to IDR integration, beta-values were first computed from the aggregated datasets. To resolve tied beta-values, a jitter factor of 0.0001 was applied using the *jitter* function in R. IDR models were then fitted to beta-values on sites with pairwise high coverage (defined as aggregated total coverage ≥ 20 in both techniques) to ensure the stability of beta-value estimation in the joint 2D space. The initial parameters for fitting the Gaussian copula mixture model (required in the *est.IDR* function) are set according to the recommended defaults in ENCODE, with $\mu = 0.1$, $\sigma = 1$, $\rho = 0.2$, and $p = 0.5$. To confirm the positivity of the IDR-identified consistent sites, binomial site-calling was also performed on the same aggregated datasets. In the end, a site was defined as orthogonally validated if it satisfied all three of the following criteria: (1) an IDR score < 0.05 , indicating high signal agreement between the two techniques; (2) a BH-adjusted P -value < 0.05 from the binomial site-calling test in technique 1; and (3) a BH-adjusted P -value < 0.05 from the same test in technique 2.

m⁶A QTL quantification and normalization

To assess the potential roles of m⁶A variation in disease pathogenesis, we performed m⁶A QTL analysis on 211 samples of the human MeRIP-seq dataset following the protocols described by Xiong *et al.* [42, 106]. All selected MeRIP-seq samples in the analysis were from wild-type conditions and used paired-end libraries. Importantly, to ensure the accuracy of findings, we only selected the human orthogonally validated m⁶A sites as the reference sites for m⁶A QTL discovery. For each of the human orthogonally validated sites, sequencing depth-normalized m⁶A modification levels were quantified as log-odds ratios (M-levels), computed as $\log[(IP / \sum(IP)) / (Input / \sum(Input))]$. Next, sites (matrix rows) with $> 20\%$ missing values (defined by zero IP + Input total coverage) across samples were removed first. Then, samples (matrix columns) with $> 80\%$ missing values across sites were removed. Subsequently, to correct for IP efficiency variability, MA-like normalization was applied as described by Zhang *et al.* [106]. Specifically, for each sample, m⁶A level profiles were compared to a pseudo reference profile (the row-wise mean across all samples), and the difference (M) and average (A) for each site were computed. A linear regression of M against A was fitted, and predicted values were subtracted from the sample profile to remove intensity-dependent biases. Sample-specific GC content biases were further corrected by fitting smoothed linear regressions with natural cubic splines across sites, following the procedure implemented in exomePeak2 [107, 108]. The corrected m⁶A levels were first normalized within each locus (row-wise) by z-score transformation and then quantile-normalized within each sample (column-wise).

m⁶A QTL detection

Latent batch effects were identified via singular value decomposition (SVD), with the number of factors estimated using the *num.sv* function from the SVA package [109, 110]. In total, 47 latent factors, together with known batch and cell type annotations, were included as covariates in the QTL analysis. Cis-m⁶A QTL discovery was performed using FastQTL v2.184 [111]. Candidate *cis*-SNPs were restricted to gene bod-

ies (exons and introns defined by Ensembl v110 and promoter regions -2000 bp upstream to $+200$ bp downstream of the transcription start site). Empirical P -values were derived from 1000 permutations per site, with beta approximation enabled ($-\text{permute } 1000 -\text{beta}$). Two stringency thresholds were applied to identify target m⁶A sites: empirical $p < 0.1$ (low) and < 0.05 (high). For each m⁶A site, nominal P -value thresholds used to decide the positive QTL lists corresponding to the two thresholds were obtained directly from the FastQTL permutation output (thresholds.txt).

Disease annotation on m⁶A QTL variants

For disease relevance, variants were annotated with GWAS [112] and ClinVar [113] data following the protocols of the previous study [114]. PLINK v.1.9.0 [115] was used to conduct the linkage disequilibrium (LD) analysis (parameters: $-\text{ld-snp-list } -\text{ld-window-kb } 1000 -\text{ld-window } 1000000 -\text{ld-window-r2 } 0.8$) with genotype data from the 1000 Genome Project [116–118]. The m⁶A QTL were mapped to these GWAS-associated diseases or traits and ClinVar curated phenotypes separately.

Orthogonal validation of m⁶A dynamics

orthogonally validated differentially methylated m⁶A sites were identified using the same IDR procedure between orthogonal techniques as described above. For each technique, biological replicates within the same cell type and perturbation condition were aggregated by summing m⁶A and total read counts per site. Only samples under matched perturbation conditions between orthogonal techniques, and sites with total coverage ≥ 20 in both techniques, were retained for integration. The differences in beta values between treated and control conditions, i.e. $(m^6A_{\text{treated}} / \text{total}_{\text{treated}}) - (m^6A_{\text{control}} / \text{total}_{\text{control}})$, were used as input for IDR analysis with the *idr* package (v.1.3) under the same parameter settings in the m⁶A site analysis. Sites with IDR < 0.05 were reported as orthogonally validated differentially methylated sites. At present, such reproducible differential m⁶A methylation sites were successfully identified only in GLORI and eTAM-seq datasets.

In-depth genomic feature annotation and analysis

Reference sites in m6AConquer were annotated in detail with transcriptomic features, RNA-binding protein (RBP) binding sites, and histone modification profiles to facilitate biological exploration and analysis. Transcriptomic features were defined as combinations of genomic regions (exons, 5' UTRs, CDS, 3' UTRs, promoters) and associated properties (length, relative position within the region, distance to 5' and 3' ends, and evolutionary conservation level), computed using the *genomeDerivedFeatures* function from the *predictiveFeatures* R package [60]. Histone modification data were obtained from ENCODE [119] for germline cell types (H1 for human, ES-E14 for mouse). m⁶A-related RBP data were derived from the m6A_ClipSeqResult datasets in the RM2Target database [120].

To analyze latent patterns among these genomic annotations, we applied a feature embedding and clustering approach conceptually similar to iMPV [121]. Dimensionality reduction was performed on all genomic annotations of the orthogonally validated m⁶A sites, first with principal component analysis (PCA) and then with uniform manifold approximation

and projection (UMAP). Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) was then applied to the reduced-dimensionality data, identifying 29 distinct clusters. The cluster annotation was created using an approach similar to scRNA-seq by identifying marker genomic features (analogous to marker genes in scRNA-seq cell clusters) for each cluster. These markers are defined as the features most important for classifying sites within or outside of a cluster, based on SHAP value analysis. For each cluster, a random forest classifier was trained to predict cluster membership of sites (in the cluster or not) using all genomic annotations as the feature input. Annotations were then assigned by interpreting the RF SHAP value diagrams for each cluster and identifying the driving features underlying each module.

Database and web interface implementation

The m6AConquer platform is built on a MySQL relational database for efficient storage and querying of base-resolution m⁶A methylation data. The web interface was developed using PHP and JavaScript, with HTML, CSS, Bootstrap, and jQuery ensuring a responsive user experience. Interactive data exploration is supported by Data Tables for tabular views and ECharts/HighCharts for dynamic visualizations. Inspired by the GTEx portal, m6AConquer provides multi-dimensional visualization of m⁶A quantification across tissues, cell types, and detection platforms [122]. A customized JBrowse instance enables intuitive navigation of genomic loci with multi-track comparison across techniques [123].

Quality control reports

We designed a multi-faceted set of quality control (QC) reports to examine sample-level data quality and its site-calling accuracy. Each report begins with a summary table of basic sample characteristics. Next, the GC content bias was evaluated by plotting sites' GC contents with m⁶A levels. Signal strength was then assessed by clustering quality in the marginal density distributions of m⁶A level. Subsequently, per-site coverage was visualized as the total and m⁶A-modified read depths. PhastCons scores were then used to compare the evolutionary conservation of methylated m⁶A sites versus unmethylated adenosine sites within ± 25 bp windows. Expression associations were evaluated by correlating M-levels with gene expression (RPKM). Genomic feature associations were analyzed using logistic regression for site classification and linear regression for methylation level prediction, incorporating known m⁶A-related features (e.g. distance to the nearest exon junction, exon length). Motif associations were quantified using ANOVA-derived R^2 values computed across 5-mers within DRACH motifs. Finally, the statistical assumptions behind different site-calling methods were benchmarked by assessing prediction accuracy on OV sites via Matthews Correlation Coefficient (MCC) and empirical fits using quantile-quantile plots. QC reports for all samples are available on the download page of the m6AConquer website.

Results

An integrated and orthogonally validated m⁶A methylome

m6AConquer is an integrated data platform that compiles quantitative m⁶A sequencing data from 1393 human and

mouse samples, obtained through ten different profiling techniques. To facilitate consistent cross-dataset analyses, we defined approximately 2.8 million reference adenine sites each for human ([2], 788 705) and mouse ([2], 761, [121]) genomes. Following this, we consistently quantified methylation ratio (as beta-value) across 11 distinct m⁶A detection techniques over the same reference sites. Our initial comparative analyses indicated moderate agreement in methylation profiles among most m⁶A detection methods, with GLORI and eTAM-seq exhibiting the highest correlation (Pearson's $r = 0.96$, Fig. 2B).

To reliably integrate data from multiple m⁶A profiling techniques, we applied the IDR framework to quantify reproducibility among orthogonal techniques, each based on distinct biochemical principles. This approach follows the methodology used in the ENCODE project for large-scale epigenetic data integration. [72]. Using this reproducibility-focused strategy, we identified 135 300 orthogonally validated m⁶A sites (IDR < 0.05) in human samples, of which 61% (82 504 sites) were further supported by at least two distinct pairs of sequencing methods. GLORI and eTAM-seq showed the highest reproducible sites (71 736 sites) among all orthogonal technique pairs (Fig. 2A). Compared to m6A-Atlas V2.0, which uses a simpler intersection-based strategy typical of established databases, our reproducibility- and orthogonality-driven approach identified a greater fraction (and number) of GLORI and eTAM Seq-confirmed sites across all thresholds (Fig. 2C). This enhanced accuracy highlights the value of our de novo quantification and statistical integration strategy, which helps mitigate method-specific artifacts that could confound simpler intersection-based approaches and those that overlook technical dependencies. Additionally, validation using these orthogonally validated sites as ground truth in benchmarks demonstrated strong performance of eTAM-seq (AUC = 0.99) and GLORI (AUC = 0.98) for site prediction (Supplementary Fig. S1). We further stratified this benchmark by DRACH motifs and other sequence contexts, and showed that our reference sites are not particularly biased toward GLORI (Supplementary Fig. S4). These results demonstrate the practical value of reproducibility-based data integration across independent m⁶A profiling techniques with different bio-chemical principles.

The m6AConquer web portal and usage

Querying and visualizing m⁶A sites for a gene of interest

m6AConquer provides gene-based queries of collected m⁶A quantitative profiles (Fig. 3A). For example, users can query and browse for m⁶A coordinates and abundances located at gene *RNF213*, which is widely recognized as the major genetic susceptibility factor for Moyamoya disease (MMD), using the gene symbol name in the search box. Detailed information of each queried m⁶A site can be accessed through the link on site ID (e.g. st2234178), such as motif sequences, supported sequencing samples and techniques, and orthogonal integration result (Fig. 3B). The m6AConquer also supports the multi-gene query by entering a list of gene symbol names. Various filters, including profiling techniques and sample cell lines or tissues, allow users to collectively explore m⁶A quantitative profiles of interest. Real-time generated m⁶A level distribution plots and heatmaps are presented as data visualization to analyze the clustering pattern of m⁶A levels among candidate genes (Fig. 3A).

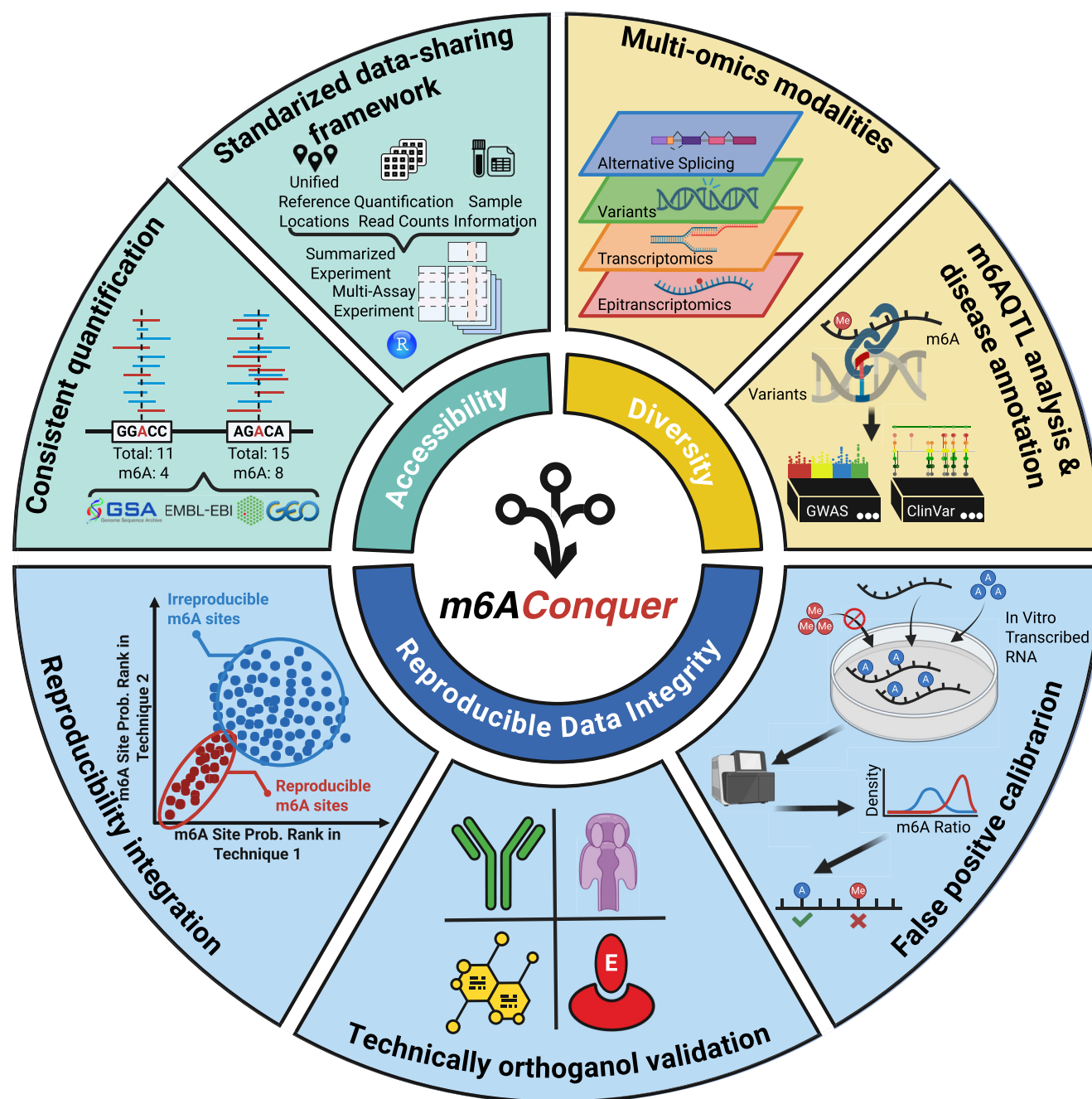


Figure 1. Overall design of m6AConquer.

Exploring reproducibility across different technologies

The m⁶A integration module contains the orthogonal validation results between pairs of techniques (Fig. 3A). Users can browse for validated m⁶A sites by the technique pairs and biological conditions (optional) of interest, retrieving the corresponding reproducible m⁶A site visualization plots and result tables. The result tables can be directly downloaded as a CSV file. The specific case of IDR integration outcomes between GLORI and eTAM-seq is presented in this manuscript (Fig. 4A). A bootstrap down-sampling simulation further indicates that model fitting and outcomes remain stable for these two orthogonal techniques, even at low sequencing depths (Supplementary Fig. S3).

Linking m⁶A variation to human disease

The m⁶A QTL module contains the m⁶A QTL mapping results and the corresponding disease annotations. Under the high stringency criteria (empirical $p < 0.05$), 6034 m⁶A QTL were identified in m6AConquer, with 428 m⁶A QTL annotated by ClinVar and 1763 m⁶A QTL annotated by GWAS. Users can filter and browse the m⁶A QTL results with target m⁶A locations, variant RS IDs, slopes and P -values (Fig. 3A). Detailed variant information and disease annotation can be accessed through the link on the site ID (Fig. 3B). To further assess the quality of our m⁶A QTL dataset, we selected m⁶A QTLs whose associated variants lie within the DRACH motif of the corresponding m⁶A sites. We observed that variants disrupting the DRACH motif have significantly more negative

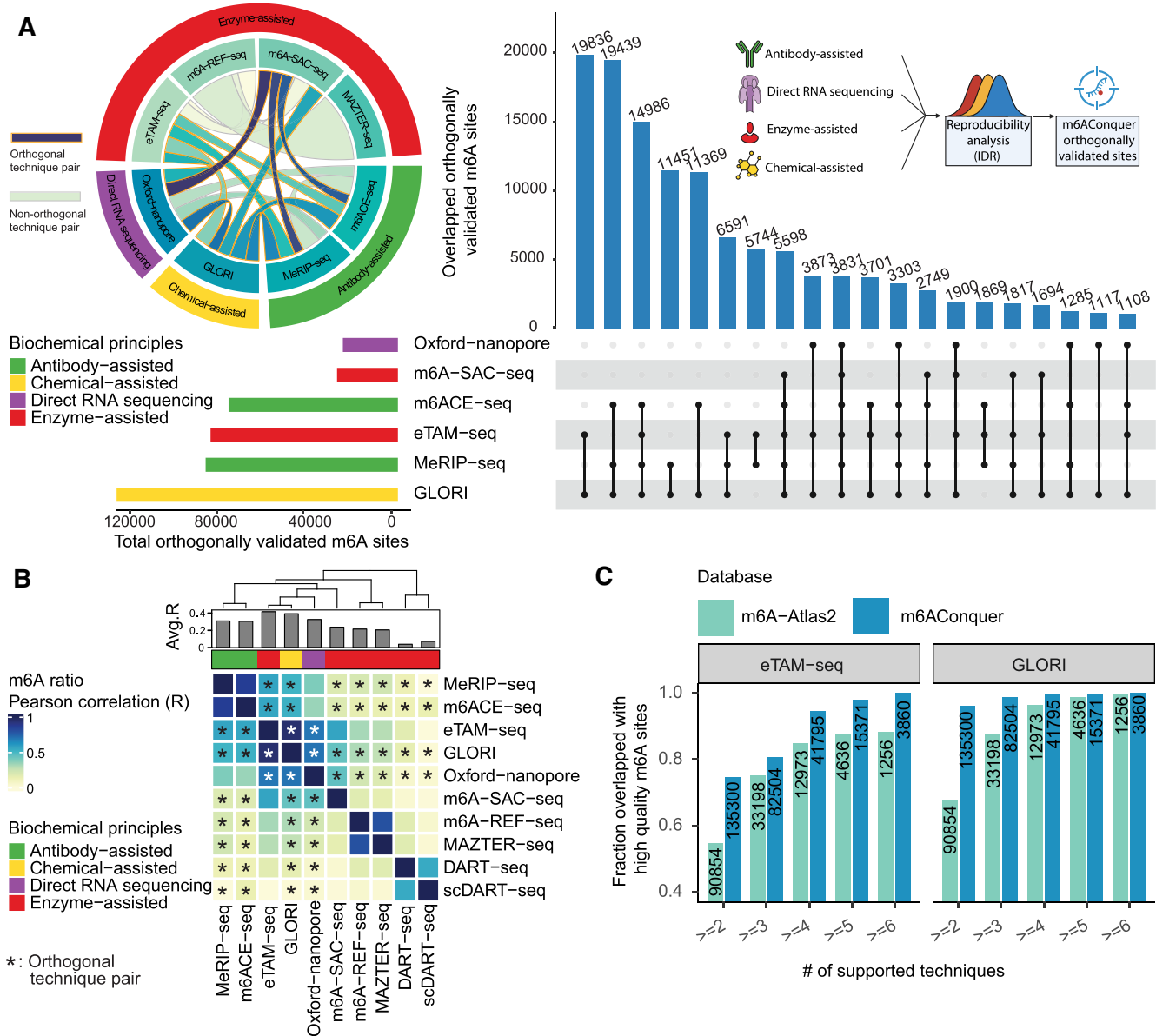


Figure 2. Overview of the consistently quantified and orthogonal validated methylome m6AConquer. **(A)** Landscape of orthogonally validated m⁶A sites. The UpSet plot (top right) visualizes the intersections of reproducible m⁶A sites (IDR < 0.05) validated across techniques with distinct biochemical principles. The horizontal bar plot (bottom left) shows the total number of unique validated sites identified with each key technology. The chord diagram (top left) illustrates the number of reproducible sites identified between any pair of techniques, with links between technically orthogonal pairs highlighted. A schematic (inset) depicts the overall validation workflow. **(B)** Pairwise Pearson correlation of m⁶A ratios (beta-values) across 10 technologies. Ratios were calculated from aggregated data within the technique. Technique categories are color-coded, and asterisks (*) denote technically orthogonal pairs. **(C)** The fraction of multi-technique supported sites verified by high-quality m⁶A sites (defined as sites also identified by eTAM-seq or GLORI). The comparison is made between sites from m6AConquer (identified by orthogonal validation) and m6A-Atlas2 (identified by simple intersection). The numbers on the bars indicate the absolute count of sites at each support level, demonstrating the superior reliability of the m6AConquer set.

m6A QTL effect sizes (slopes) compared to those that do not (t-test $p = 0.00775$). It is worth noting that, in our dataset, all observed slopes for DRACH-disrupting variants were negative (Supplementary Fig. S2).

Convenient and multi-modal data download

m6AConquer provides multi-modal data for download (Fig. 3C). Firstly, our web server allows the export of both single-omics and multi-omics data-sharing frameworks

for omics quantification data. Secondly, the orthogonally validated m⁶A sites are also available in both R Genomic Ranges files (in RDS format) or tabular text files (in CSV and BED format). Both combined and cell-line-specific integration are provided. Lastly, users can also retrieve and download the quality control reports for each sample. To facilitate ease of use, the m6AConquer web portal is accompanied by extensive documentation, step-by-step tutorials, and detailed guides on its data-sharing frameworks.

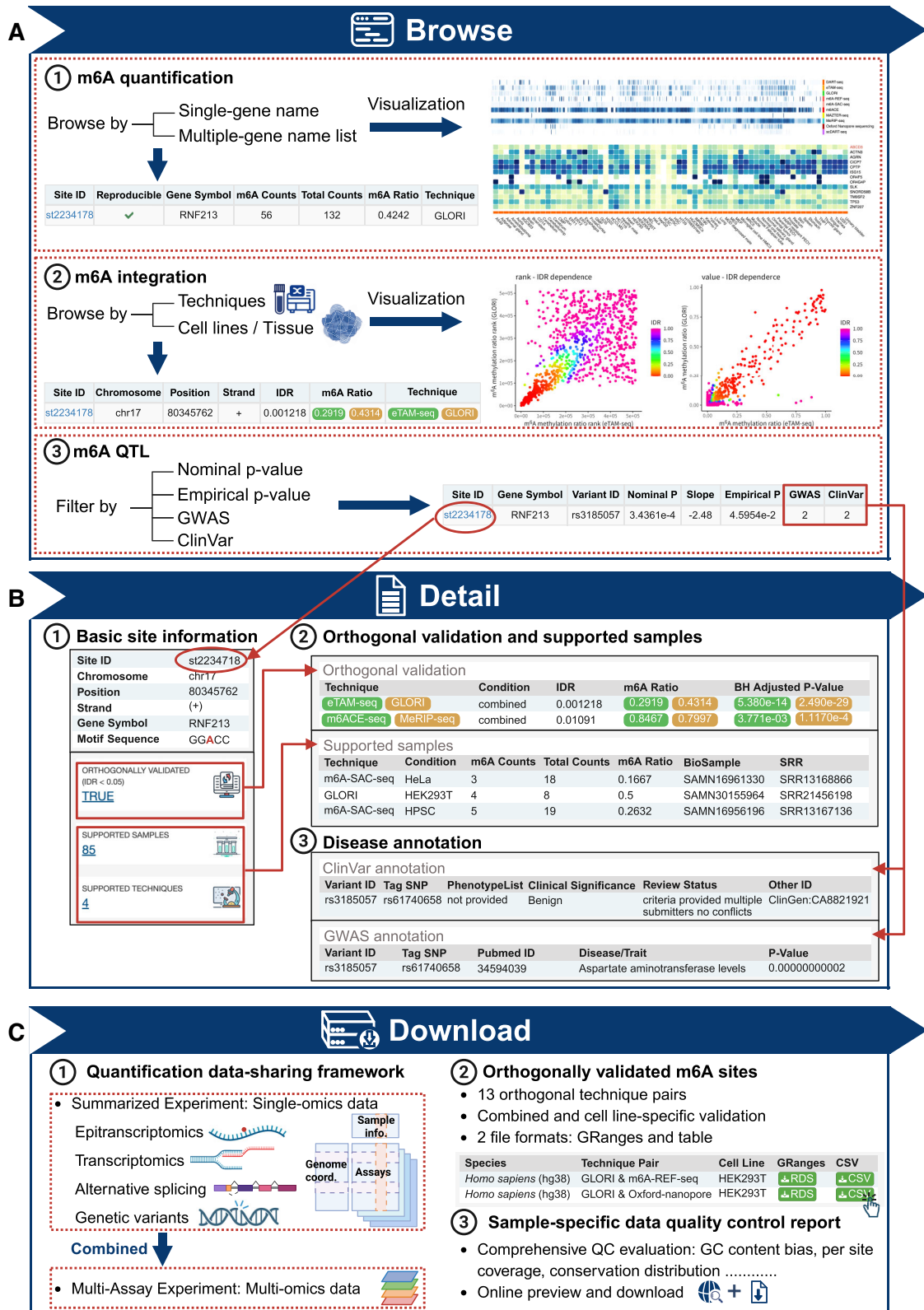


Figure 3. Interactive query interface of the m6AConquer website. (A) m6AConquer provides data browsing for m⁶A quantification, m⁶A integration, and m⁶A QTL data with various filters and data visualization. (B) Clickable Site IDs (e.g. st2234178) access detailed information, including Motif sequences (e.g. "TGACT"), Supporting techniques/samples, Orthogonal reproducibility status, and Disease annotation. (C) Download module in m6AConquer contains standardized data-sharing framework (single-omics and multi-omics), orthogonally validated m⁶A sites, and a sample-specific data quality control report.

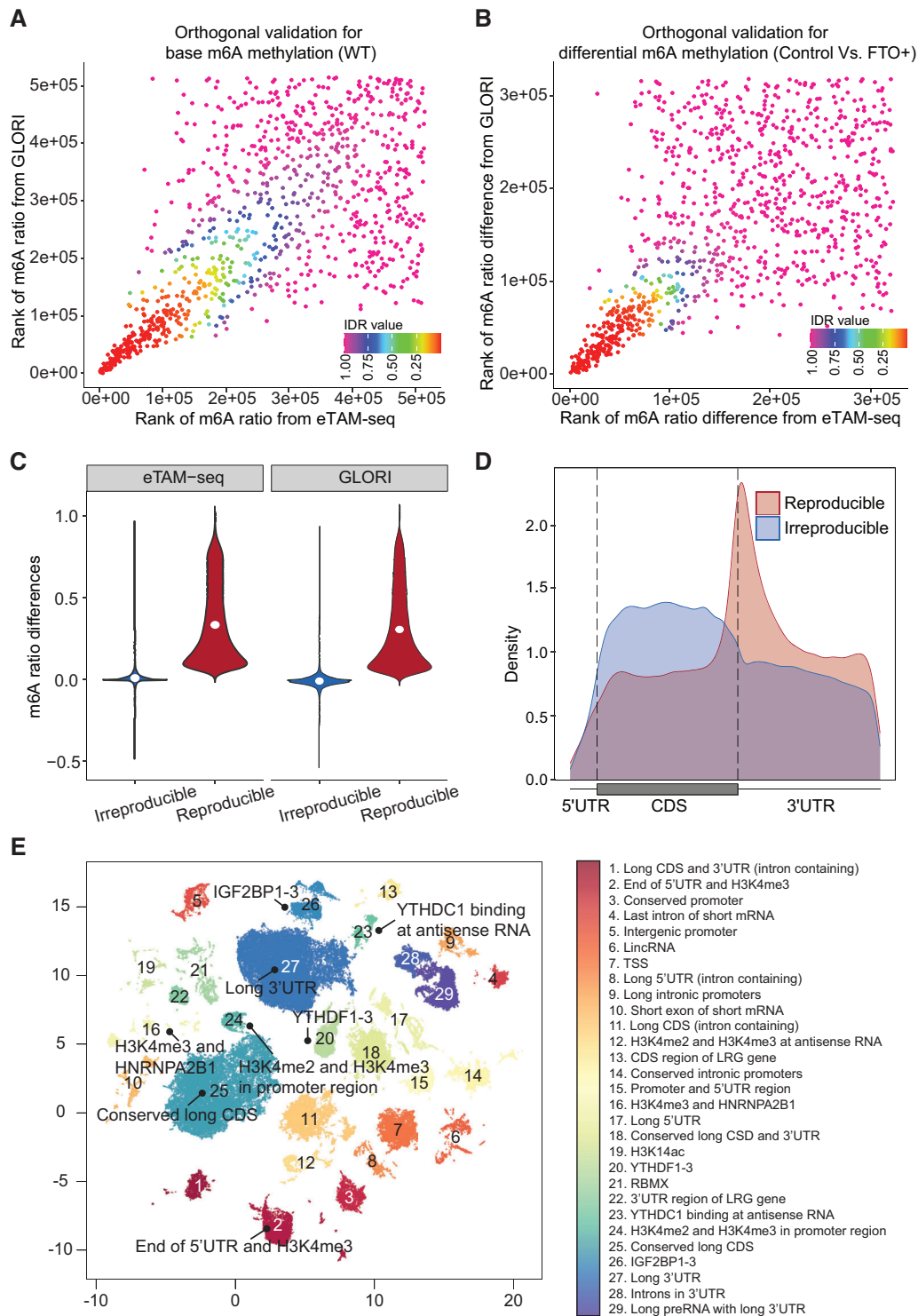


Figure 4. Orthogonal validation reveals robust and functionally stratified m⁶A sites. **(A)** Rank-rank scatter plot of m⁶A ratio (beta-value) between eTAM-seq and GLORI, colored by IDR. Axes represent the site ranks from each technique. **(B)** Rank-rank scatter plot of m⁶A ratio differences between control and FTO-treated in-vitro RNA samples. Axes represent the ranks of differences from each technique. Scatter plots in A and B are downsampled to 1000 points for visualization clarity. **(C)** Comparing m⁶A ratio differences for reproducible vs. irreproducible differentially methylated sites as determined by IDR < 0.05 in Fig. 4B. Reproducible sites show larger differences. **(D)** Transcript topology profile of the orthogonally validated FTO responsive sites ($n = 62,002$), demonstrating enrichment near stop codons. **(E)** Unsupervised clustering of orthogonally validated m⁶A sites in human based on their genomic features. A two-dimensional UMAP projection was generated from 184 genomic annotations associated with each orthogonally validated m⁶A site. HDBSCAN was then applied to the UMAP embedding to identify distinct clusters. The plot displays 29 identified clusters, with cluster labels indicating the driving genomic annotations for each cluster. The cluster label was obtained from SHAP value analysis of random forest classifiers, which predicted whether a site belongs to a given cluster using 184 genomic annotations. As a result, the driving factors identified for each cluster include transcript features (e.g. long 3'UTRs), promoter-associated chromatin marks (e.g. H3K4me3), and RNA-binding protein (RBP) interaction hubs (e.g. IGF2BP1-3). This stratification reveals the diverse regulatory landscapes where m⁶A modification occurs and suggests that distinct functional roles are governed by the local genomic context.

Case Study 1: Identifying orthogonally validated differential methylation

To demonstrate its utility in studying m⁶A dynamics, m6AConquer supports the identification of orthogonally validated, differentially methylated sites. As a case study, we investigated changes in m6A methylation following the in-vitro demethylase treatment of RNA with FTO, using data from the technically independent GLORI and eTAM-seq methods. We first calculated the change in methylation levels (beta-value differences) for sites with sufficient coverage (≥ 20 reads). Applying our IDR statistical framework to these differential methylation effect sizes (Fig. 4B), we successfully identified 62 002 reproducible sites of differential methylation (IDR < 0.05). Interestingly, these orthogonally validated FTO-responsive sites showed a distinct enrichment near the stop codon (Fig. 4D), consistent with established basal m⁶A topology. The validated differential sites also exhibited substantially larger changes in methylation ratio compared to irreproducible sites (Fig. 4C).

Case Study 2: Uncovering regulatory contexts of m⁶A sites via unsupervised clustering

To showcase the potential of the rich set of genomic annotations curated in m6AConquer, we performed an unsupervised clustering of genome features on the orthogonally validated m⁶A sites in human. All necessary data for this analysis, including site coordinates and 184 genomic features spanning transcript topology, RBP binding, and histone modifications, were exported directly from m6AConquer's annotation dataset. Dimensionality reduction via UMAP revealed clear patterns of functional organization, stratifying the m⁶A sites into 29 distinct clusters based on their genomic context (Fig. 4E).

Interestingly, we further found that each cluster was driven by a distinct combination of features, potentially uncovering their diverse regulatory roles. For instance, Cluster 2, characterized by proximity to the 5'UTR end and co-occurrence with H3K4me3, suggests a role in transcriptional initiation. In contrast, Cluster 27 represents sites within long 3'UTRs, implicating them in mRNA stability. The analysis also identified key "m⁶A hubs" defined by the co-localization of specific reader proteins, such as YTHDF1-3 (Cluster 20) and IGF2BP1-3 (Cluster 26). It also identified m⁶A sites bound by histone modifications H3K4me2 and H3K4me3, which are associated with the promoters of genes containing evolutionarily conserved CDS (Cluster 24 and Cluster 25). This case study demonstrates that the multi-layered, detailed genomic annotation data within m6AConquer provide a useful resource for dissecting the complex regulatory landscapes of m⁶A.

Discussion

The rapid progress in epitranscriptomic sequencing has created both immense opportunities and a critical challenge: the lack of reproducibility and coherent integration of data from disparate sources. In this work, we developed m6AConquer to address this challenge by replacing simple data aggregation with consistent re-quantification and integration strategies grounded in reproducibility. Our approach leverages the rigorous IDR framework to assess concordance across orthogonal techniques, thereby establishing a high-confidence refer-

ence m⁶A methylome. This integration strategy effectively filters out technique-specific artifacts and spurious signals, a conclusion supported by its observed improvement in accuracy over a simple intersection-based m⁶A database and by the high performance of absolute quantitative techniques (eTAM-seq and GLORI) evaluated in our curated benchmark dataset.

A central goal of m6AConquer project is to facilitate the exploration of m⁶A's functional roles. By providing matched multi-omics data within a unified data-sharing framework, the platform supports researchers to connect methylation events to downstream consequences. For example, our m⁶A QTL analysis of m⁶A sites validated by orthogonal methods illustrates how users can reliably connect genetic variation to RNA methylation, generating testable hypotheses about m⁶A dysregulation in human disease.

Beyond functional genomics, our consistent data-sharing framework offers a highly valuable resource to the computational research community. While challenges such as batch effects, technical artifacts, and distributional shifts can confound tool and model evaluations, as seen in the ENCODE imputation challenge [124], m6AConquer directly mitigates these issues through its rigorous, consistent, and analysis-ready data curation. Crucially, our platform provides complete and unbiased quantification matrices that include both methylated and unmethylated sites. Together, these features address the data sparsity and selection bias inherent in "positive-call-only" databases, offering an ideal, high quality substrate for training and benchmarking generalizable AI models.

The m6AConquer project currently faces several limitations. First, it does not fully eliminate heterogeneity arising from the upstream processing steps of different m⁶A sequencing methods, as the initial pipelines before quantification and site-calling (e.g. trimming and read alignment) still follow the standard protocols from each technique's original publication. Second, although it applies consistent quantification procedures and IVT calibration wherever possible, it cannot fully remove technique-specific artifacts caused by detection limitations, such as the low sequencing depth in ONT-based methods and the highly skewed profiles generated by non-absolute quantification techniques. Future studies should therefore focus on automating data standardization through reproducible software workflows implemented using Docker containers and on developing additional ML tools that leverage IVT control samples to calibrate false positives across diverse sequencing technologies.

In conclusion, m6AConquer offers the scientific community a rigorously curated dataset and a transparent strategy for m⁶A analysis. While the current work prioritizes quality over quantity, future developments aim to achieve both. The dataset will grow in parallel with the field's emphasis on high-quality, orthogonal data from absolute quantification methods (e.g. GLORI, eTAM-seq) and emerging chemical conversion-based techniques such as m6A-CAM-seq [125]. As these results accumulate, their integration will enable the construction of a far more detailed map of validated m⁶A profiles, allowing for high-resolution analysis of m⁶A dynamics across diverse cell types and perturbations. By providing an consistently quantified and orthogonally validated atlas of the m⁶A methylome, we hope this resource will serve as a practical *ground truth* for computational benchmarking and enhance the reliability of continued advances in RNA methylation research.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

D.J.R. holds the position of Executive Editor for *Nucleic Acids Research* and has not peer reviewed or made any editorial decisions for this paper.

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Data availability

The m6AConquer database is freely available to all users without any login requirement at <https://rnamd.org/m6aconquer/>. The source code for processing raw data in m6AConquer is provided at GitHub (<https://github.com/XichenZhao0223/m6AConquer-Data-Processing>) with an accompanying Zenodo archive (DOI: 10.5281/zenodo.17169694). The source code for the binomial/BBmix site-calling is available in the reproducible R package OmixM6A at GitHub (<https://github.com/ZW-xjtlu/OmixM6A>) and archived at Zenodo (DOI:10.5281/zenodo.13337113). The curated data matrices in SummarizedExperiment and MultiAssayExperiment formats are available for batch download from the database website.

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